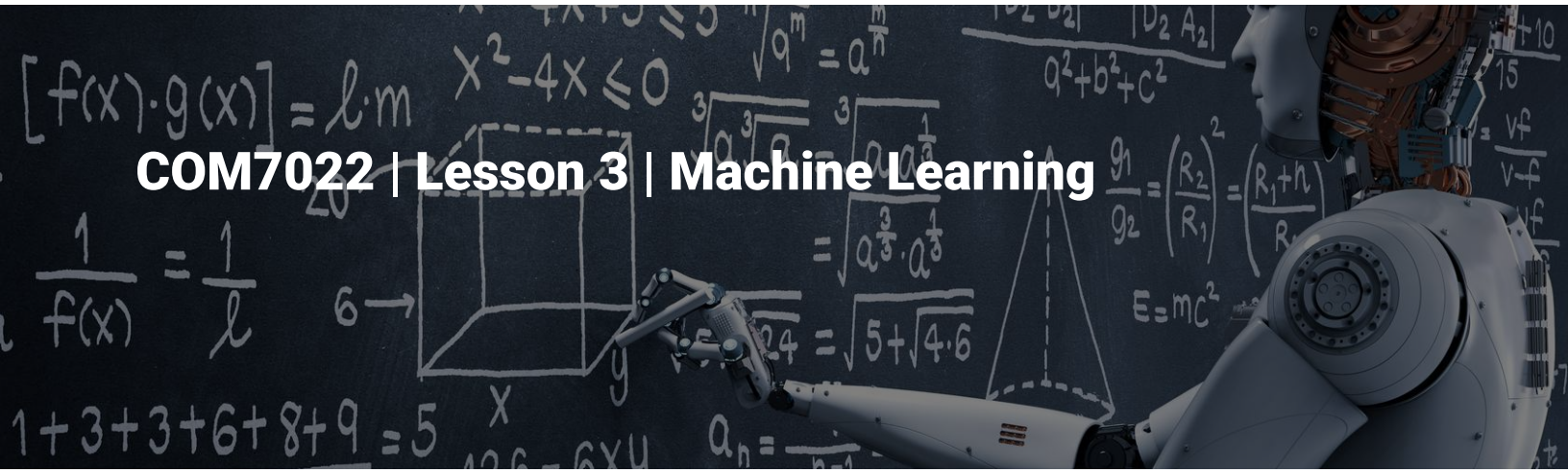


COM7022 | Lesson 3 | Machine Learning

Machine Learning (COM7022)

Lesson 3: Regression

Cover image by Phonlamai Photo from Shutterstock.com

- **Lesson introduction**
- **Before you get started**
- **Topic 1: Simple linear regression**
- **Topic 2: Multiple linear regression**
- **Topic 3: Polynomial regression**
- **Topic 4: LASSO regression**
- **Knowledge check**

Lesson summary

Reading List

Lesson introduction

Welcome to Lesson 3

LESSON INTRODUCTION | LEARNING OUTCOMES | ACCESSIBILITY GUIDANCE

Welcome to Lesson 3!

We are entering an exciting lesson in the module. Lesson 3 will give us the first glimpse into the wonderful world of data science by introducing our first machine-learning algorithm: regression. In this lesson, we will explore regression, finding the relationships between attributes and outputs (referred to as dependent and independent variables in some literature).

Now that we are moving on to some hands-on practical topics, let us explore them alongside an active Jupyter Notebook to go through the practical elements together.



Computing

At Arden University

LESSON INTRODUCTION | LEARNING OUTCOMES | ACCESSIBILITY GUIDANCE

By the end of this lesson, you will:

- Understand the workings of **simple linear regression** and how to implement and visualise the output
- Understand the additional benefit of **multiple linear regression** when dealing with multiple attribute datasets
- Know how to apply **polynomial regression** and its differences from linear regression models
- Explore the implementation of **LASSO regression**

Based on the above, Lesson 3 aligns with all three learning objectives:

1. Demonstrate a practical understanding of the characteristics of various machine-learning methods and their applications in data analysis.
2. Select and apply appropriate machine-learning techniques for a given scenario.
3. Critically evaluate the result from the application of machine-learning techniques and propose improvements.

The following are the topics we will cover in this lesson:

- Topic 1: Simple linear regression
- Topic 2: Multiple linear regression
- Topic 3: Polynomial regression
- Topic 4: Least



We have worked across teams at Arden University to try and create an accessible and inclusive experience within the learning content. We are continuously assessing the accessibility of our lessons to make improvements and enhancements based on testing and feedback.

This lesson may include screenshots of code and software used. Please be aware that these may not always have a text alternative, and that other features such as size and colour may not be accessible.

If you experience any issues with accessing the learning content throughout this module, there are several support services in place to help you.

- Arden University has a dedicated Inclusion Services Team who can help you with getting adjustments and equipment.
- The Library Portal also has information on the accessibility tools that can help you access resources: Accessibility Tools – Library Portal
- If you are struggling with accessing a specific part of the content e.g. a video resource that is missing captions, please speak to your Lecturer who may be able to advise on an alternative resource in the interim to ensure you are able to get the learning you need to progress with the lesson.

Continue to the Before you get started section

Before you get started

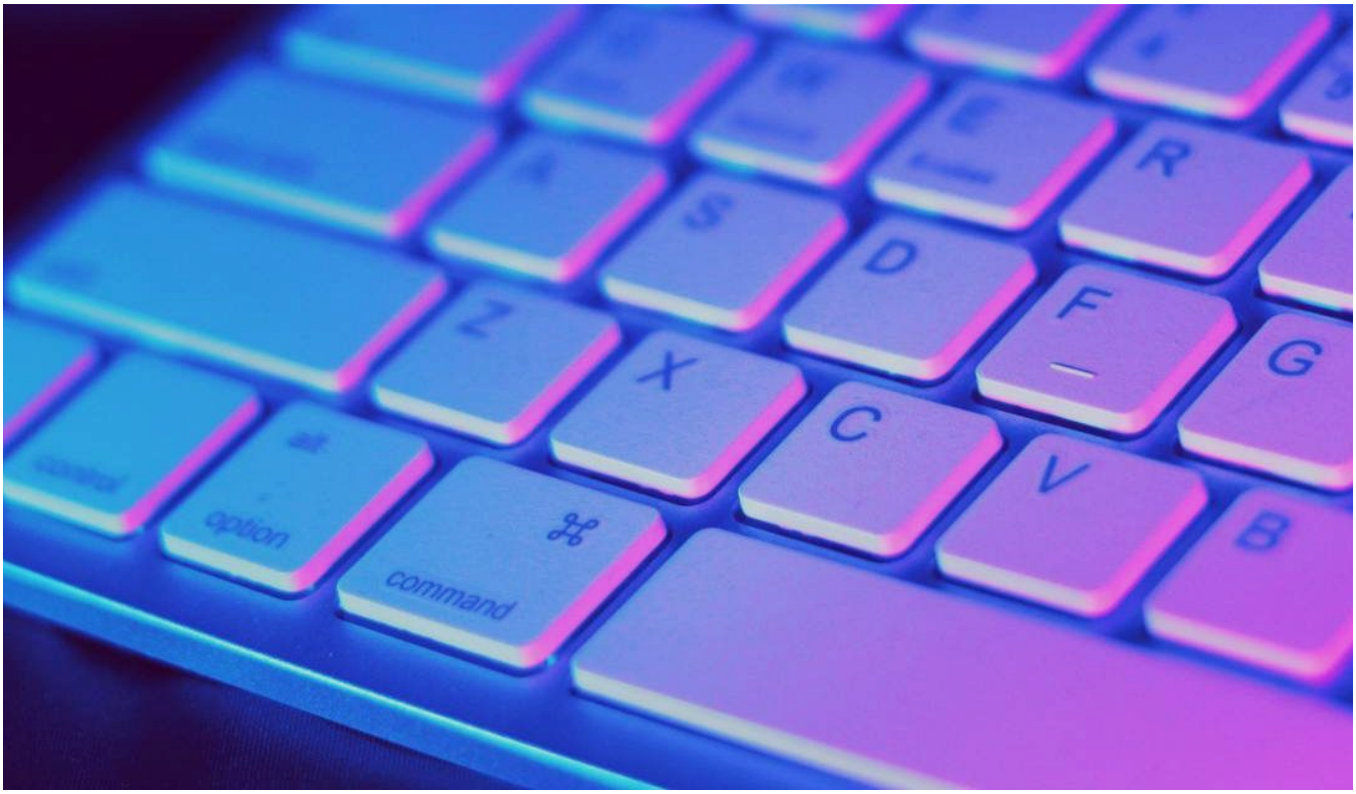


Image from Articulate Rise 360 Content Library.

When working through the practical elements in this lesson, many of you will be tempted to “cut and paste” the code directly from the text onto Jupyter Notebook and run them. However, creating the Notebook from scratch and practising coding can be very beneficial. This will allow you to debug any syntax errors you might encounter and learn from these mistakes. This opportunity will also help you better understand the program’s logic flow and syntax. Most importantly, it will make your learning experience more interactive and hands-on.

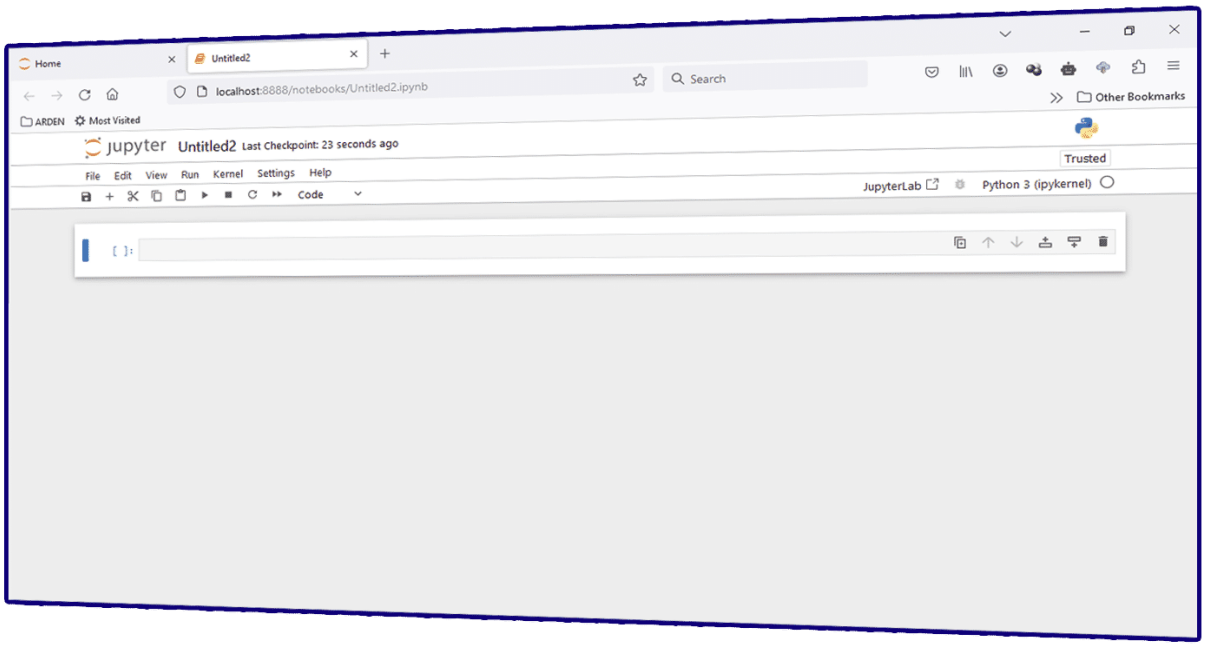


Image by Arden University.

Continue

Did you know?



Image from Articulate Rise 360 Content Library.

What are the top ten data scientist job titles, and what do they entail?

Read: 12 Data Science Job Titles — Which Role Is Right for You? by Metwalli and Urwin

Continue

What do you think "prompt engineering" is?



Image from Articulate Rise 360 Content Library.

Continue

With the rise of generative AI tools such as ChatGPT, a new skill known as “prompt engineering” has emerged. This skill involves designing prompts that will maximise the benefits of generative AI tools (McKinsey & Company, 2022).

Continue to Topic 1

Topic 1: Simple linear regression

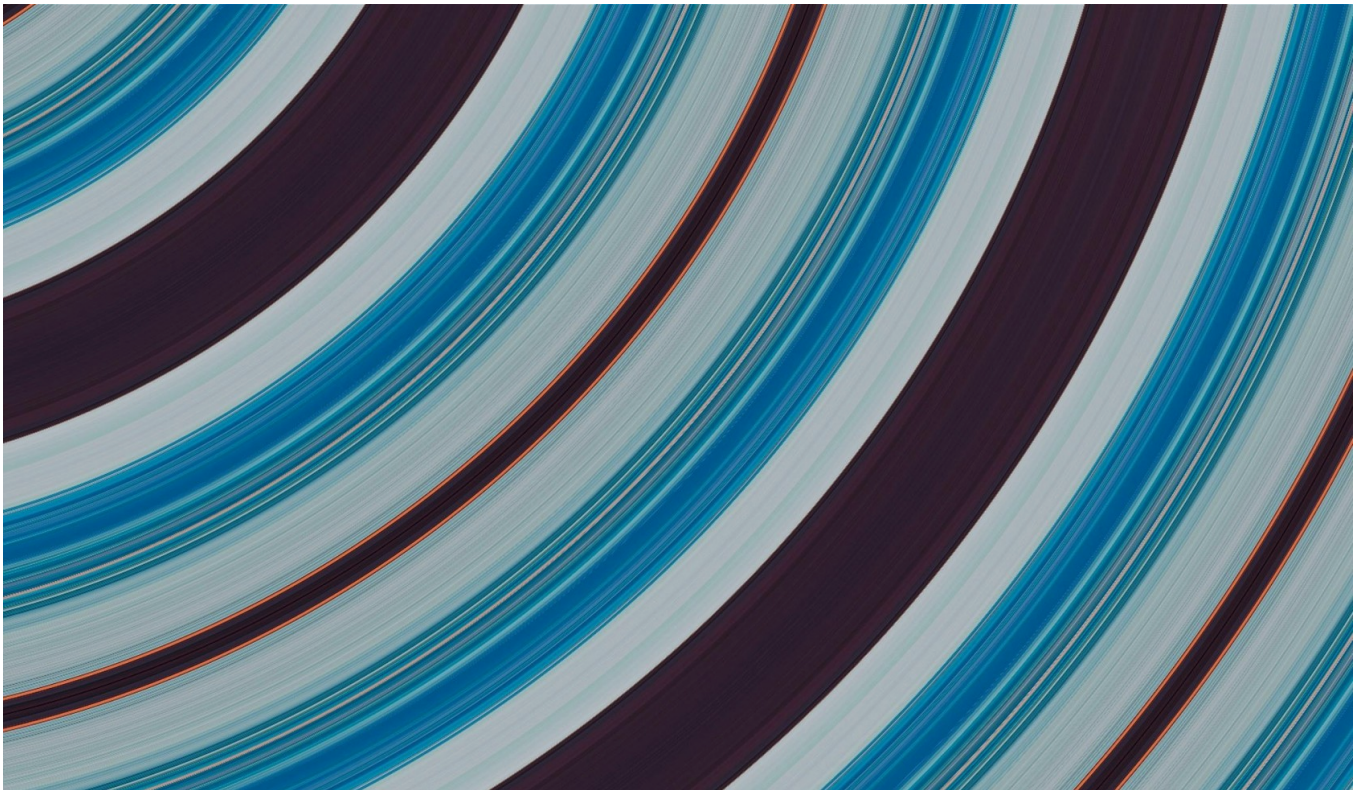


Image from Articulate Rise 360 Content Library.

Linear regression (LR) is one of the most common types of prediction algorithms used in the industry. In this topic, we will look at simple linear regression (SLR) applied to a single independent variable/attribute/feature versus a single dependent variable/response variable/outcome variable.

Although we might not encounter an SLR model in our day-to-day life as data scientists, we can visualise the relationship between features and response variables by studying SLR.

When we say “linear”, we should remember a straight line on a graph with:

$$y = mx + c \qquad \text{(equation 1)}$$

Which can be visualised as:

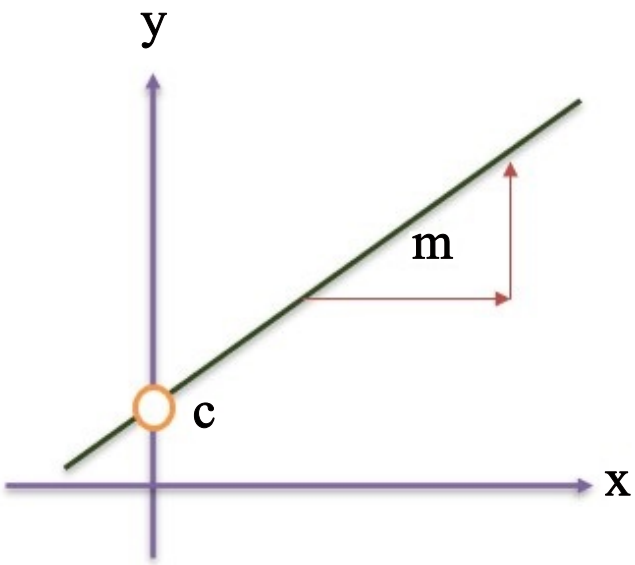


Figure 3.02. Graph of a straight line by Arden University. (Accessible text).

Where:

- *y* = dependent variable/response variable/outcome variable
- *x* = independent variable/attribute/features
- *m* = gradient of the line
- *c* = y-intercept/threshold

Continue

Let us look at linear regression in action with the following example:

Dataset = Salary versus Experience

Business problem:

HR has asked you to develop a simple prediction model to predict the estimated salary of new employees based on the dataset they have on the current employees within the company.

Using the dataset provided, we can visualise the dataset as follows:

Years Experience	Salary(£)
1.1	39343
1.3	46205
1.5	37731
2	43525
2.2	30801
2.9	56642
3	60150
3.2	54445
3.2	64445
3.7	57189
3.9	63218
4	55794
4	56957
4.1	57081
4.5	61111
4.9	67938
5.1	66029
5.3	83088
5.9	81363
6	93940
6.8	91738
7.1	98273
7.9	101302
8.2	113812
8.7	109431
9	105582
9.5	116969
9.6	112635
10.3	122391
10.5	121872

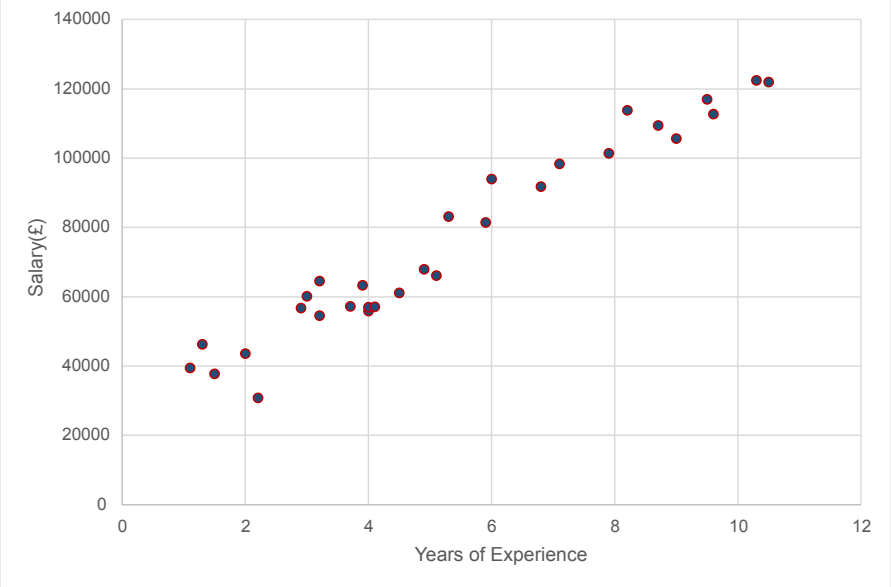


Figure 3.03. Years Of Experience vs Salary (£) by Arden University. (Accessible text). (Excel file).

Without doing any calculations, the intuition of LR is to plot a best-fit line across the data points that will allow us to make use of the plotted line to predict y (Salary(£)) based on x (YearsOfExperience) as shown below:

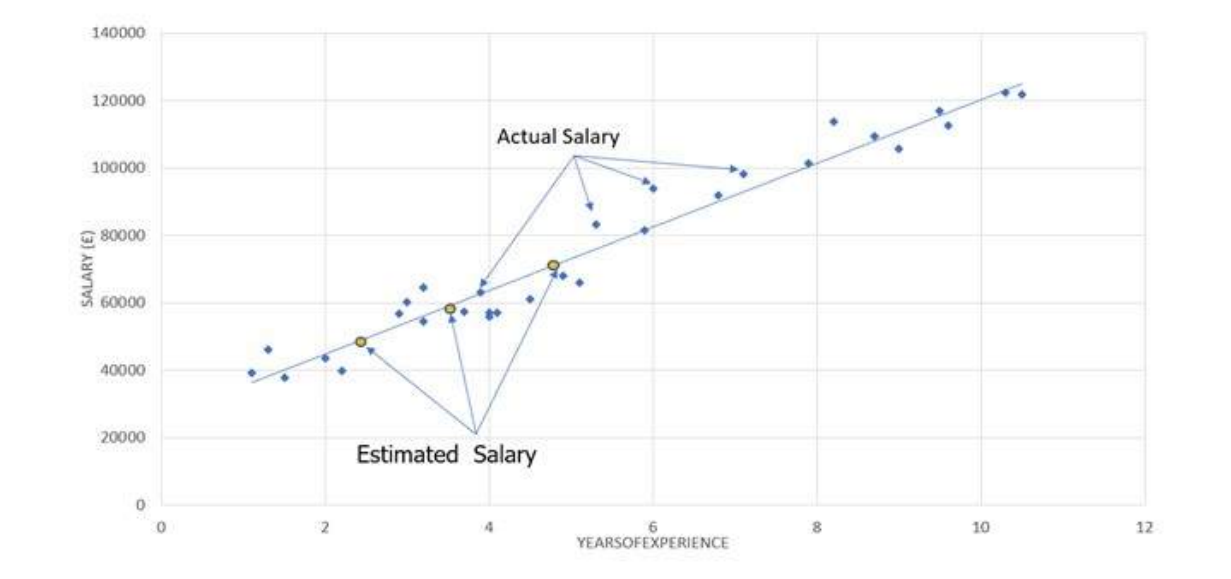


Figure 3.04. Years Of Experience vs Salary (£) – Linear regression by Arden University.

Continue



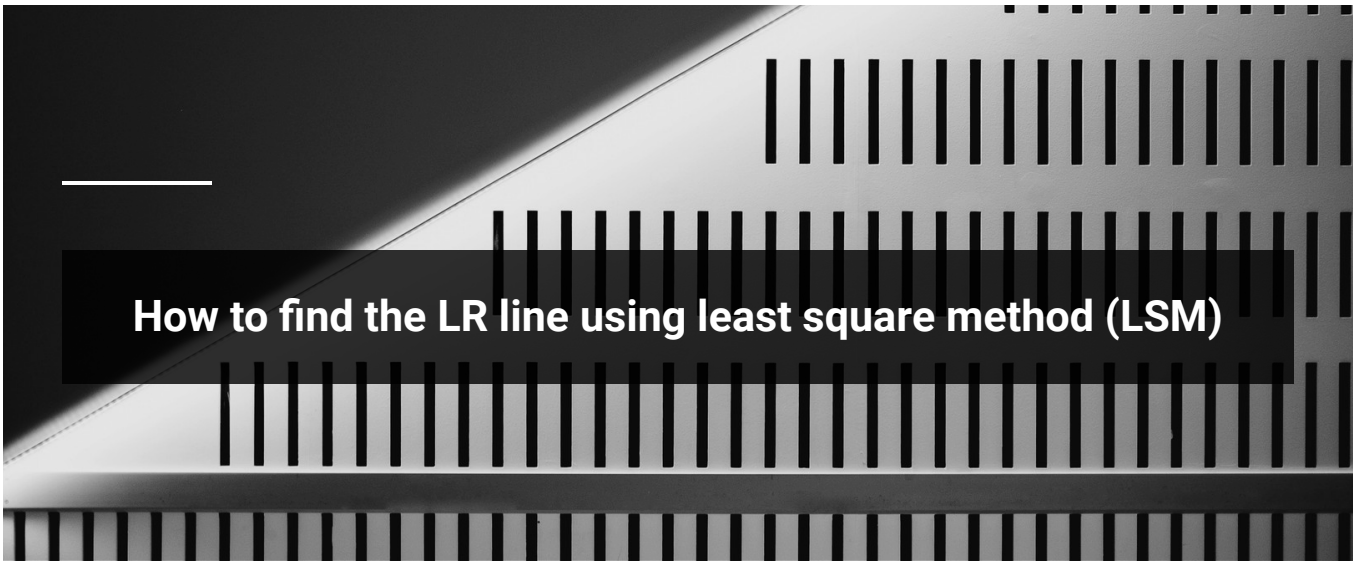


Image from Articulate Rise 360 Content Library.

Step 1: Calculate the regression line:

We know that:

$\hat{y} = mx + c$ (equation 2)

Where:

- $\hat{y}$ = estimated salary
- x = Years of experience
- m = gradient/coefficient (in Python)
- c = y-intercept

$$m = \frac{\sum(x - \overline{x})(y - \overline{y})}{\sum(x - \overline{x})^2} \qquad (equation\ 3)$$

Figure 3.06. Gradient equation by Arden University.

Where:

- $\bar{x}$ = mean of x
- $\bar{y}$ = estimated salary

To find c, substitute *m* into (equation 2); *x* with $\bar{x}$ as the regression line must pass through the mean of *x*.

Continue

Step 2: Calculate the performance measure (R^2)

Ideally, R^2 should be within 90–99% to be optimal. If not, it is essential either to increase the dataset instances or to remove any outliers within the dataset.

$$R^2 = \frac{sum\ of\ distance\ (estimated - mean)^2}{sum\ of\ distance\ (actual - mean)^2}$$

$$R^2 = \frac{\sum (\widehat{y} - \overline{y})^2}{\sum (y - \overline{y})^2} \qquad (equation\ 4)$$

Figure 3.07. R^2 by Arden University.

Continue

A practical example of LR in Jupyter Notebook:

The program is shown below using the Years Of Experience vs Salary (£) dataset:

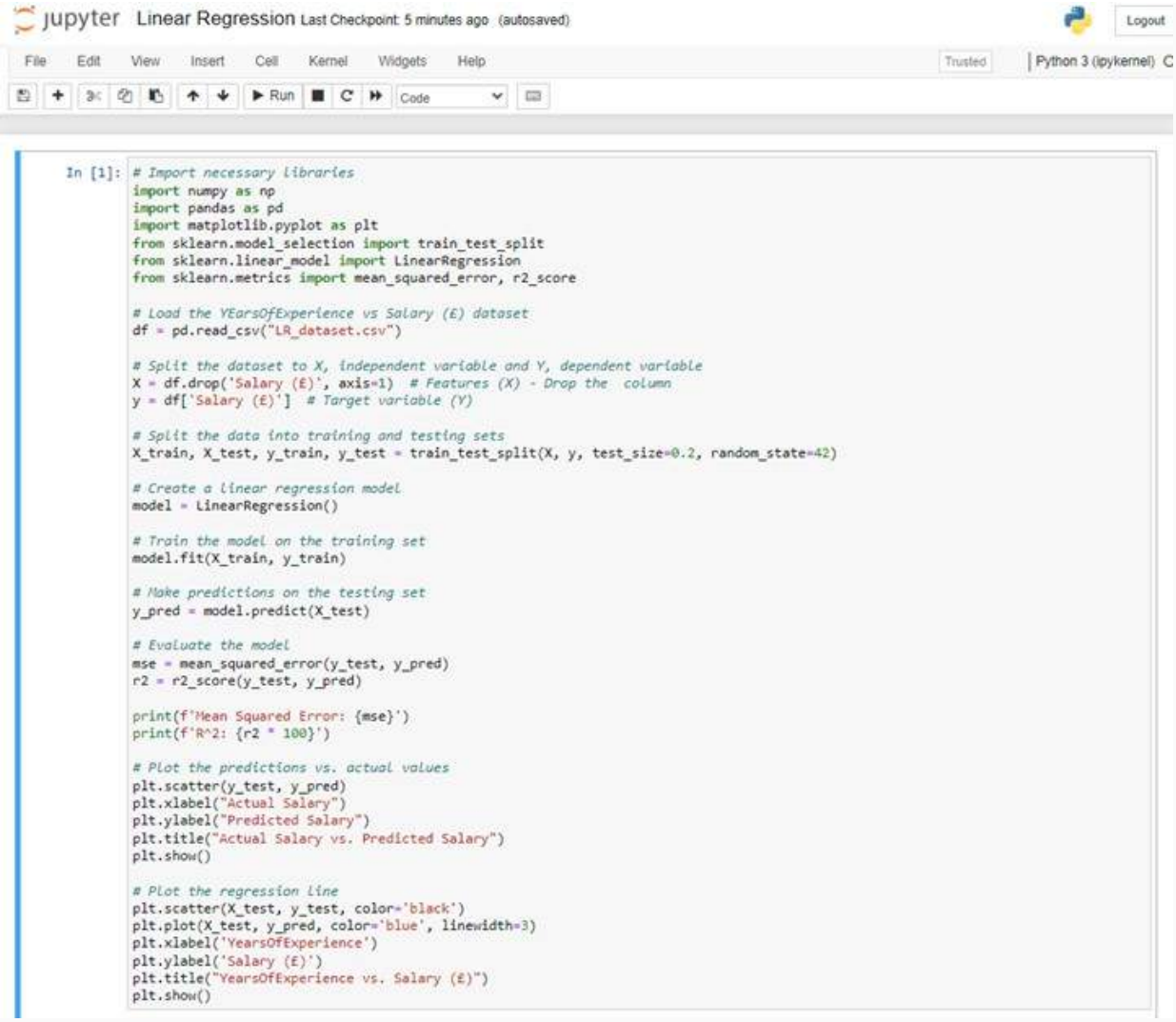


Figure 3.08. Linear regression model in Jupyter Notebook by Arden University. (Accessible code).

The outputs of the program are as follows:

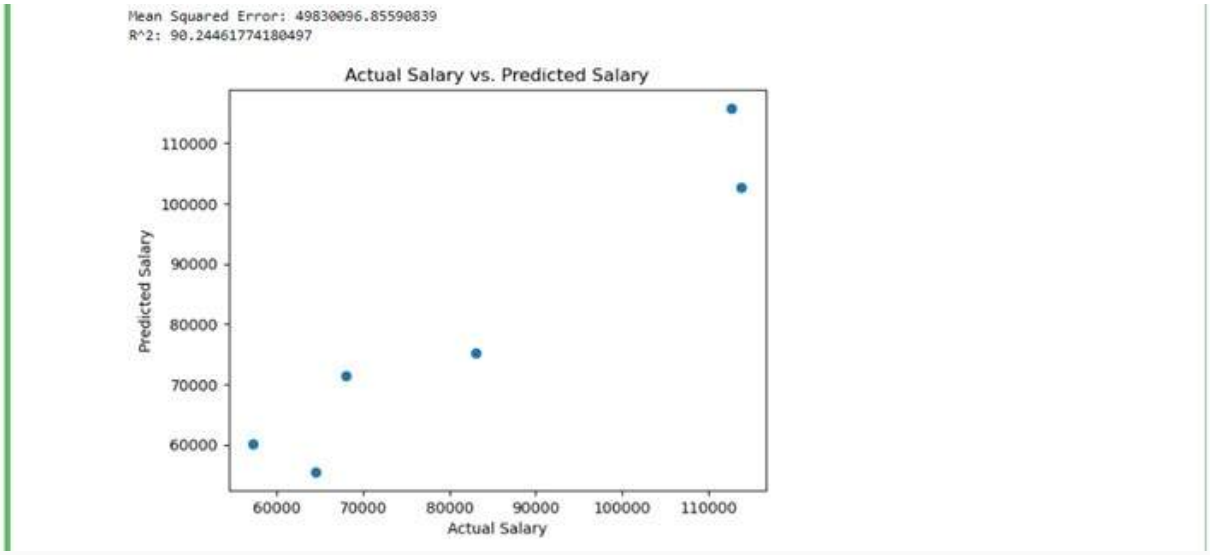


Figure 3.09. Linear regression model outputs in Jupyter Notebook by Arden University.

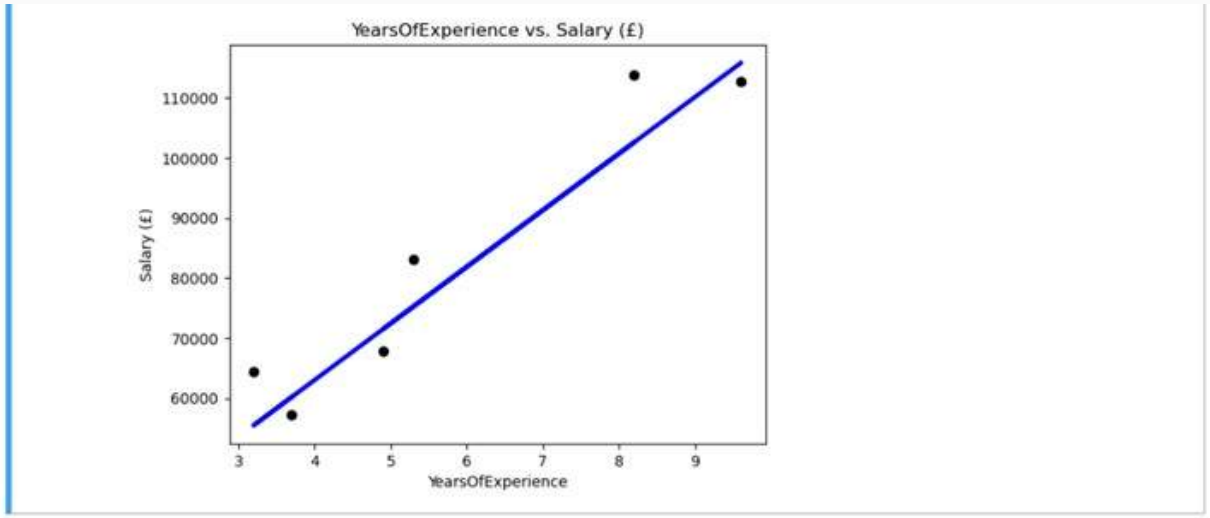
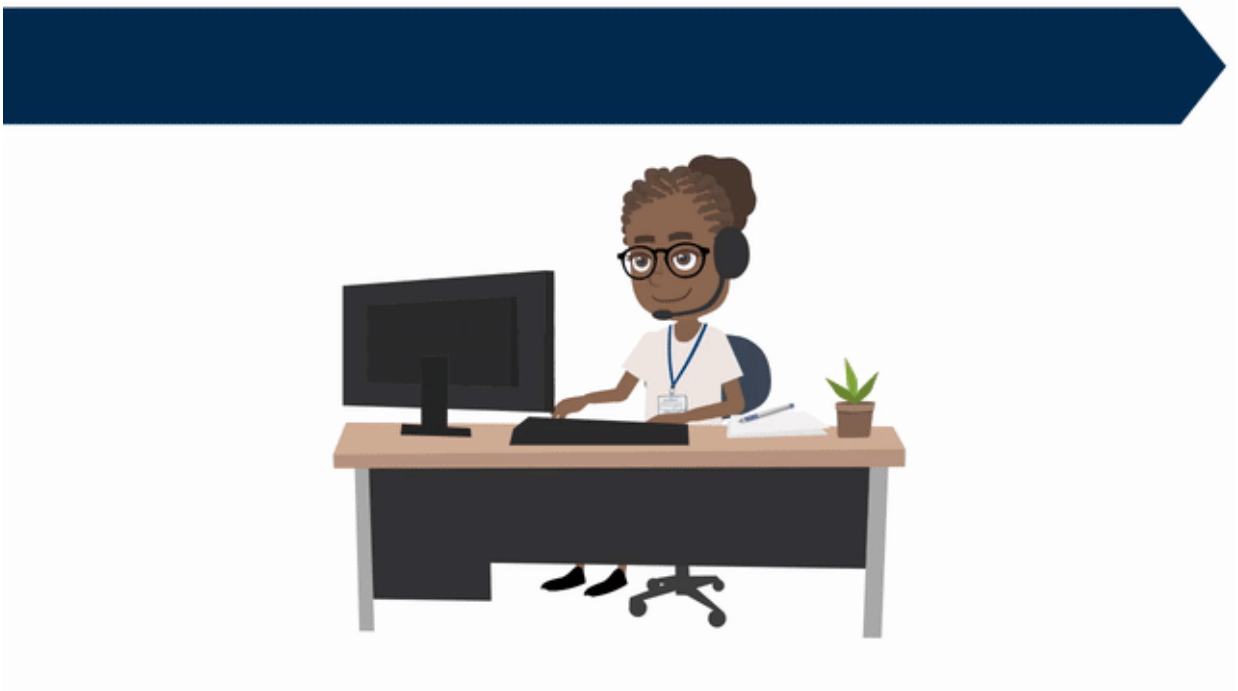


Figure 3.10. Linear regression line for the simulation in Jupyter Notebook by Arden University.

Continue

Topic 1 - Activity 1

Examine the relationship between independent and dependent variables



From the example in this section, we can clearly see that when the YearsOfExperience increases, the Salary (£) increases. We call this a positive correlation relationship. With this knowledge, please determine the relationship between X and Y in the following figures:

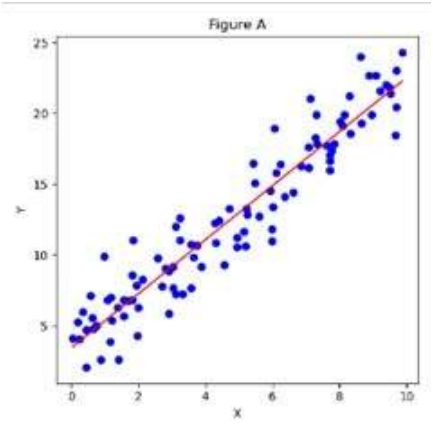


Figure A

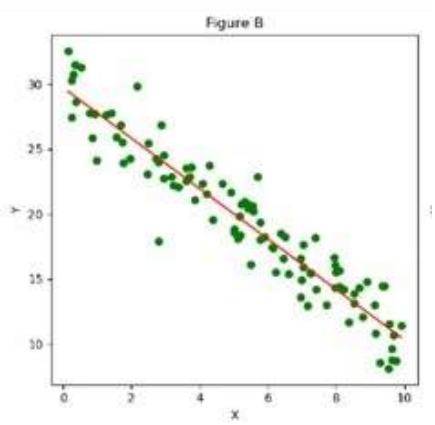


Figure B

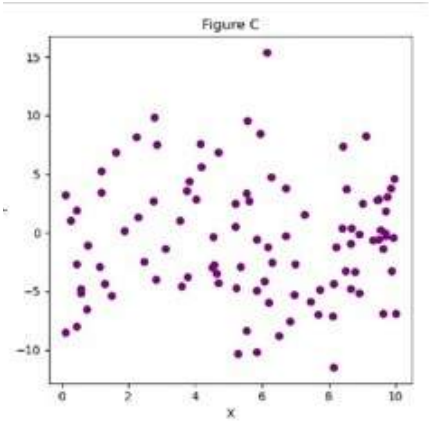


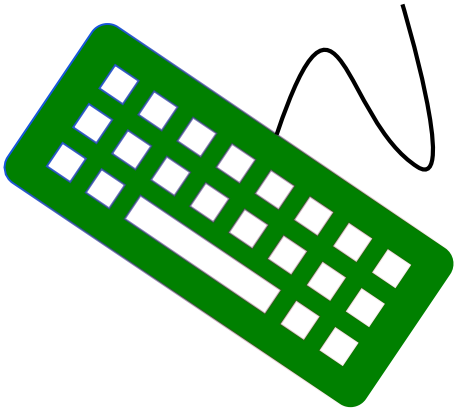
Figure C

Continue

Topic 1 - Activity 2

Hands-on exercise to develop your own linear regression model

Using the example in this section, find a suitable dataset and create a regression model.



Continue

Topic 1 - Activity 3

Reading



- 9.3: Statistical Approaches (pp. 730-739) in Chapter 9: Anomaly Detection from Introduction to Data Mining, Global Edition by Tan *et al*.
- Chapter 8: Simple Linear Regression, Chapter 9: Multiple Regression and Model Building, and Chapter 13: Logistic Regression from Data Mining and Predictive Analytics by Larose and Larose.

Continue to Topic 2

Topic 2: Multiple linear regression



Image from Articulate Rise 360 Content Library.

In the previous topic, we explored SLR with one independent variable and one dependent variable, and we used the regression line to predict the dependent variable values. In this topic, let us further explore LR when you have more than one independent variable.

In simple LR, we have the following relationship:

$$x \rightarrow y$$

However, in multiple linear regression (MLR), we are looking at multiple independent variables, as illustrated in the following figure:

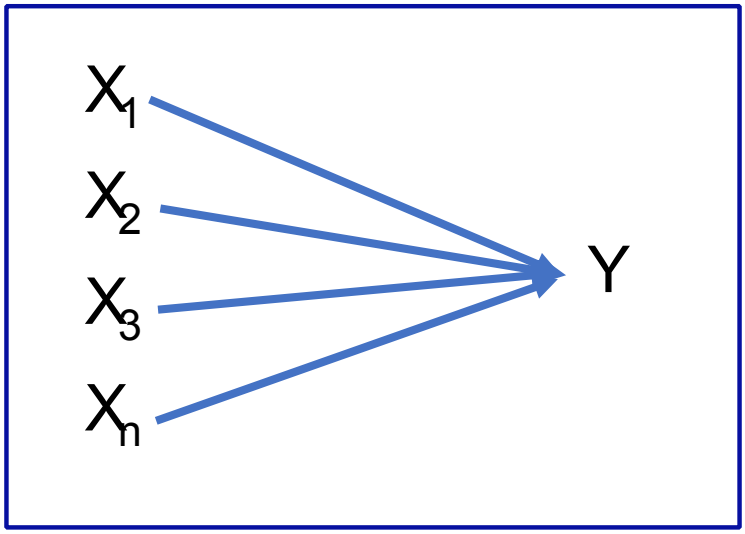


Figure 3.13. Relationship between multiples X and Y by Arden University. (Accessible text).

We can visualise the regression plane with two independent variables with MLR. (Note: we can only visualise three-dimensional space.)

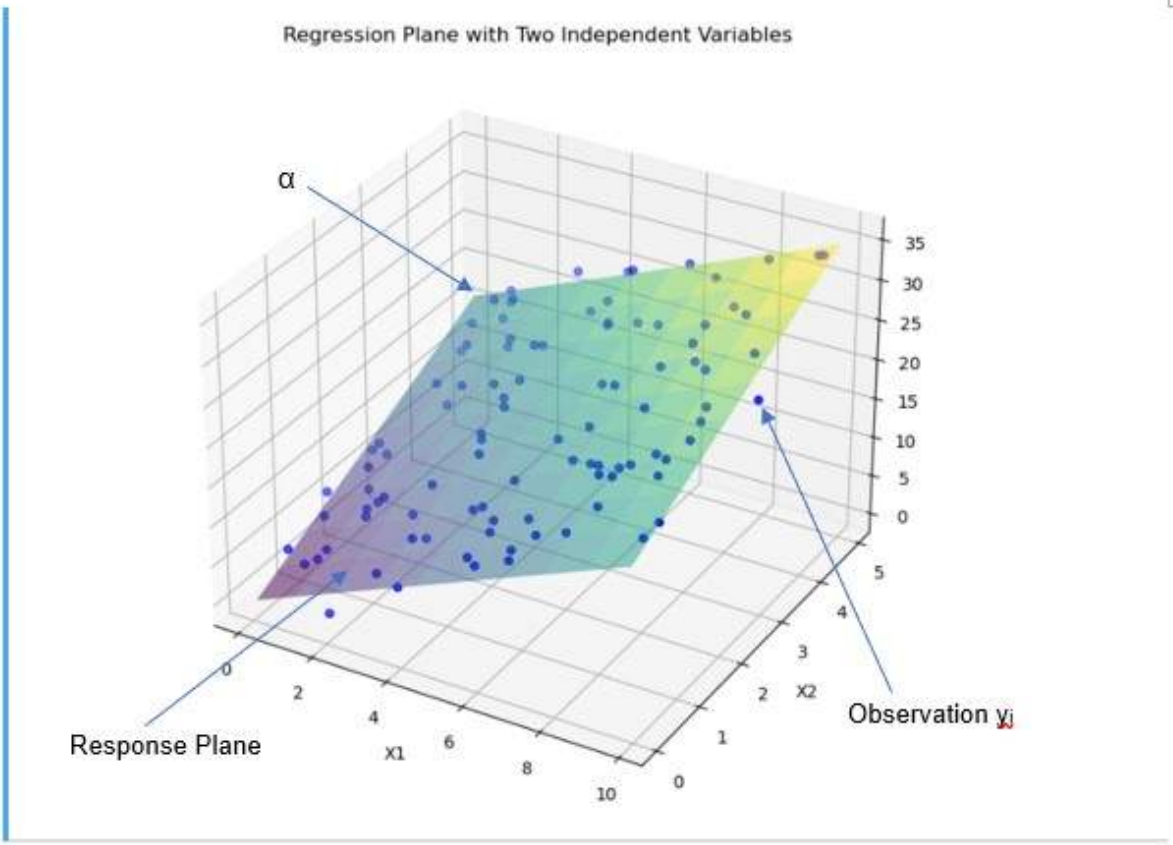


Figure 3.14. MLR modelling in Jupyter Notebook by Arden University.

How to find the MLR line using least square method

We know that:

$$\hat{y} = \beta_1 x_1 + \beta_2 x_2 + \alpha \qquad \textbf{(equation 5)}$$

Where:

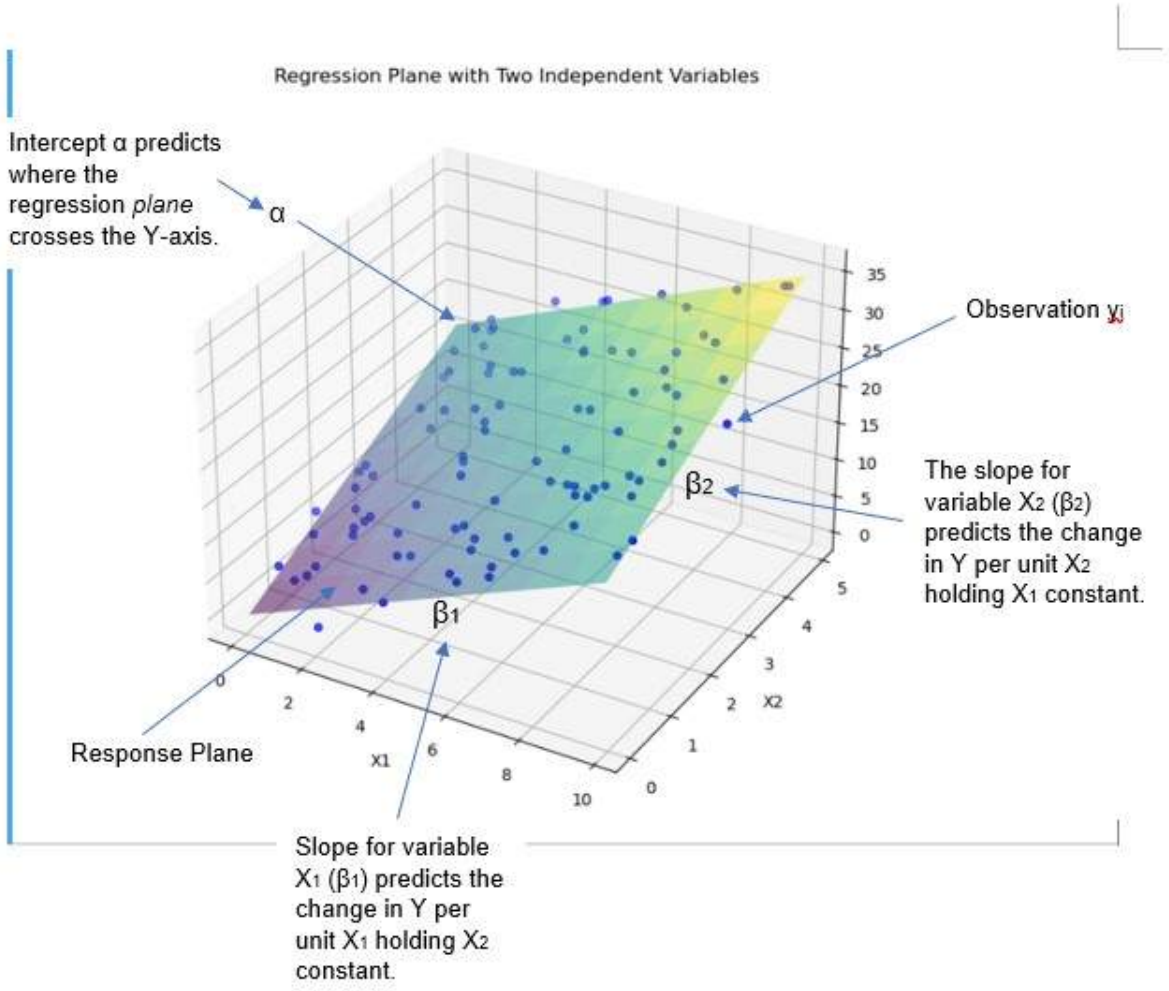
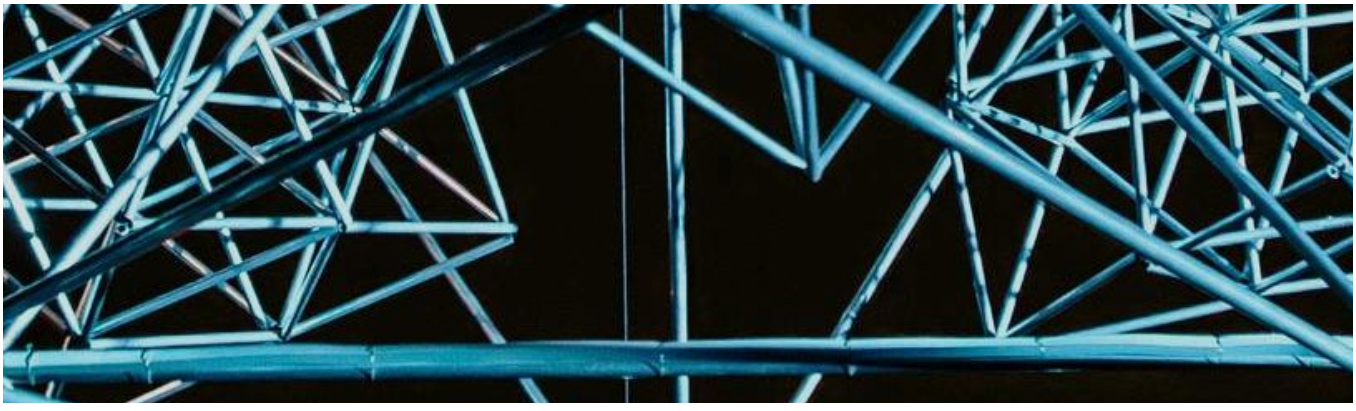


Figure 3.16. MLR equation model in Jupyter Notebook by Arden University. (Accessible text).

Continue



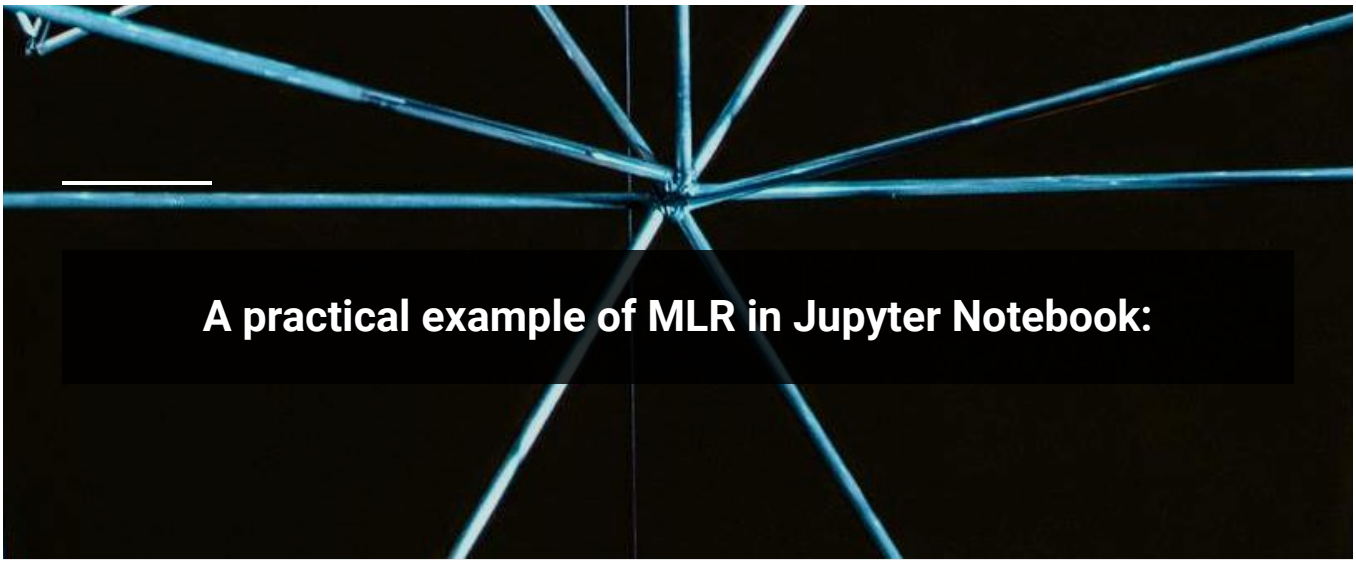


Image from Articulate Rise 360 Content Library.

Business statement: Investors would like to find out, based on a company's R&D spend, admin and marketing spend, what the profit of a new startup will be.

The following are the steps to build the MLR model:

- **Step 1:** Import the libraries
- **Step 2:** Read in the dataset
- **Step 3:** Split the dataset into X = independent variables, Y = dependent variable
- **Step 4:** Splitting the dataset into the training set and the test set
- **Step 5:** Fitting MLR to the training set
- **Step 6:** Predicting the test set results
- **Step 7:** Building the optimal model using backward elimination

Backward elimination or **backward deletion** is the reverse process. All the independent variables are entered into the equation first and each is deleted one at a time if it does not contribute to the regression equation.

The data we will be using is as follows:

R&D_spend	Admin	Marketing_spend	Profit
165349.2	136897.8	471784.1	192261.8
162597.7	151377.6	443898.5	191792.1
153441.5	101145.6	407934.5	191050.4
144372.4	118671.9	383199.6	182902
142107.3	91391.77	366168.4	166187.9
131876.9	100814.7	362861.4	156991.1
134615.5	145198.9	127716.8	156122.5
130298.1	145530.1	323876.7	155752.6
120542.5	148719	311613.3	152211.8
123334.9	108679.2	304981.6	149760
101913.1	110594.1	229161	146122
103672	91790.61	249744.6	144259.4
93863.75	127320.4	249839.4	141585.5
91992.39	135495.1	252664.9	134307.4
119943.2	156547.4	256512.9	132602.7
114523.6	122616.8	261776.2	129917
78013.11	121597.6	264346.1	126992.9
94657.16	145077.6	282574.3	125370.4
91749.16	114175.8	294919.6	124266.9
86419.7	153514.1	125304	122776.9
76253.86	113867.3	298664.5	118474
78389.47	153773.4	299737.3	111313

Figure 3.17. MLR dataset by Arden University. (Accessible text). (Excel file).

The program code for the model is:

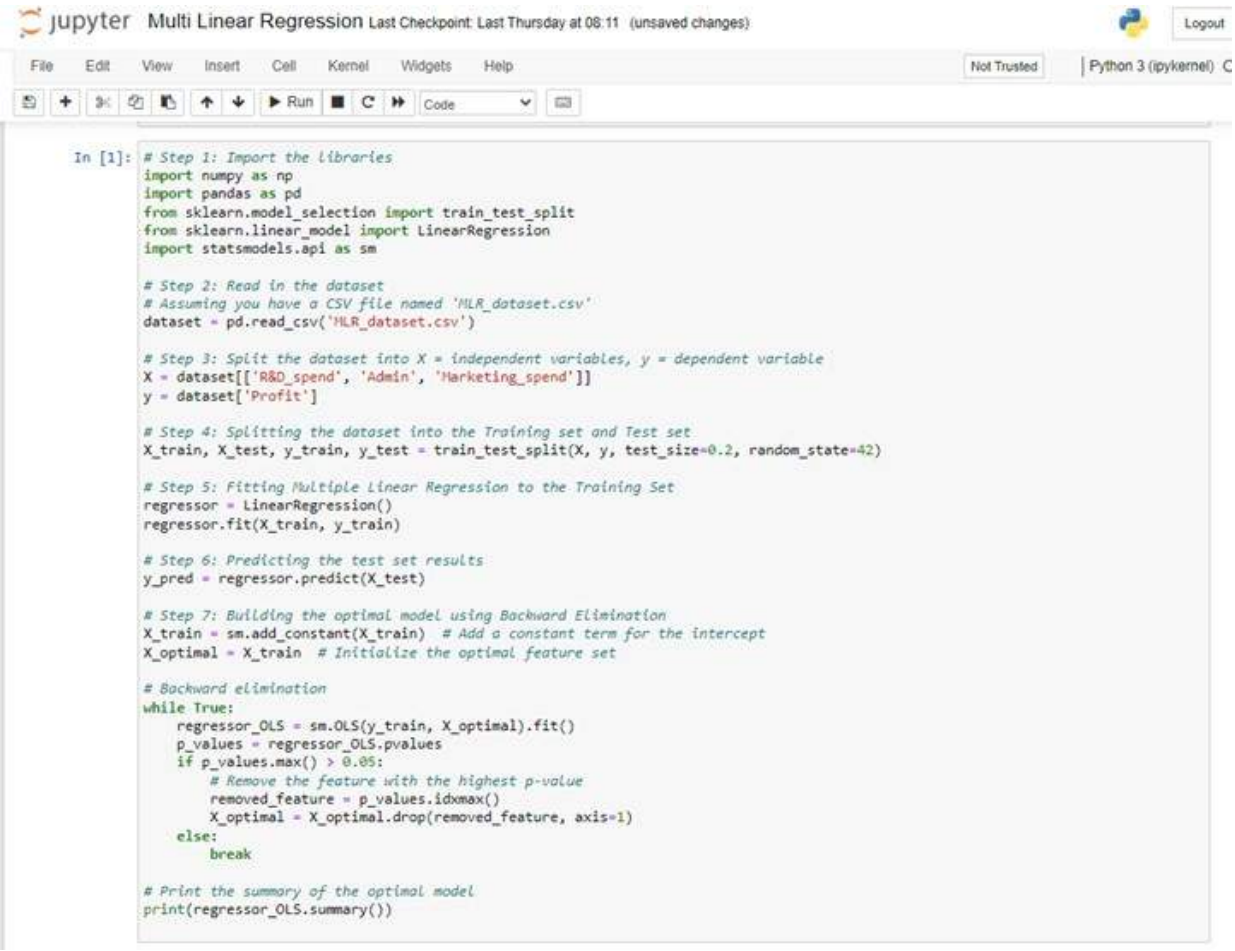


Figure 3.18. MLR program code in Jupyter Notebook by Arden University. (Accessible code).

The output:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                 Profit   R-squared:                       0.893
Model:                            OLS   Adj. R-squared:                  0.878
Method:                 Least Squares   F-statistic:                     58.64
Date:                Mon, 11 Dec 2023   Prob (F-statistic):           1.57e-07
Time:                        08:37:57   Log-Likelihood:                -174.88
No. Observations:                  17   AIC:                             355.8
Df Residuals:                      14   BIC:                             358.3
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const       8.798e+04   1.65e+04      5.342      0.000    5.27e+04    1.23e+05
R&D_spend      0.7499      0.080      9.370      0.000       0.578       0.921
Admin         -0.2174      0.092     -2.369      0.033      -0.414      -0.021
==============================================================================
Omnibus:                        1.557   Durbin-Watson:                   2.708
Prob(Omnibus):                  0.459   Jarque-Bera (JB):                0.972
Skew:                           0.234   Prob(JB):                        0.615
Kurtosis:                       1.926   Cond. No.                     1.45e+06
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.45e+06. This might indicate that there are
strong multicollinearity or other numerical problems.
```

Figure 3.19. MLR program output in Jupyter Notebook by Arden University.

The “best” model selection depends on various factors, and it often involves trade-offs between simplicity and explanatory power. In the context of MLR with backward elimination, the process involves iteratively removing independent variables based on their p-values until all remaining variables have p-values below a certain significance level (commonly 0.05).

Continue

In the example above, the loop removes features one at a time based on their *p*-values. The final model will contain the features that contribute significantly to explaining the variance in the dependent variable.



To interpret the results, look at the summary of regressor_OLS.summary(). The “P>|t|” column in the summary provides the *p*-values for each coefficient. You can consider the corresponding variable statistically significant if the *p*-value is less than your chosen significance level (commonly 0.05).

For example, if the summary shows that R&D_spend has a *p*-value less than 0.05 and Admin and Marketing_spend have *p*-values greater than 0.05, you might choose a model with only R&D_spend as the best model.



The decision might also involve considering practical or domain-specific knowledge. For instance, you might choose to keep a variable even if its p-value is slightly above 0.05 if it is theoretically important or if removing it negatively impacts the interpretability of the model.

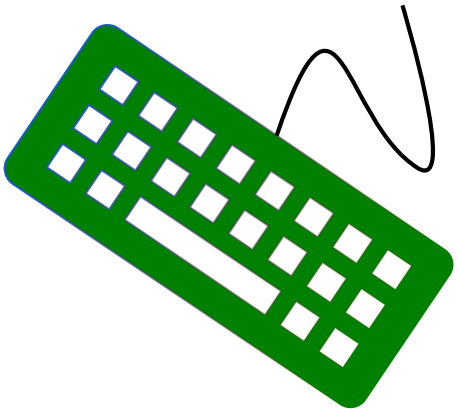
Continue

Topic 2 - Activity 1

Activity 1: Hands-on exercise to develop your own MLR model

Create an MLR model using 60 randomly generated data points, two independent variables and one dependent variable. Plot the 3-D plane of best fit and display the equation of the plane plus the R squared value.

From the model created above, how could you improve its accuracy? Implement the improvements and run the model again. Have the changes improved the model's accuracy?



Continue

Topic 2 - Activity 2

Reading



- 13.7 ,13.8 and 13.9 (pp. 370-382) in Chapter 13: Logistic Regression from Data Mining and Predictive Analytics by Larose and Larose.
- Pages 113-124 of Chapter 4: Analysis of Business Intelligence and Big Data Adoption in Organizations in Business Intelligence and Big Data: Drivers of Organizational Success by Olszak.

Continue to Topic 3

Topic 3: Polynomial regression



Image from Articulate Rise 360 Content Library.

So far in this lesson, we have been dealing with non-polynomial regression (i.e., there is no exponent or power on the independent variable, *X*, in all our datasets). However, life always throws you a curve ball, as do the datasets you will encounter professionally. For example, the figure below shows the clear difference between linear and polynomial datasets.

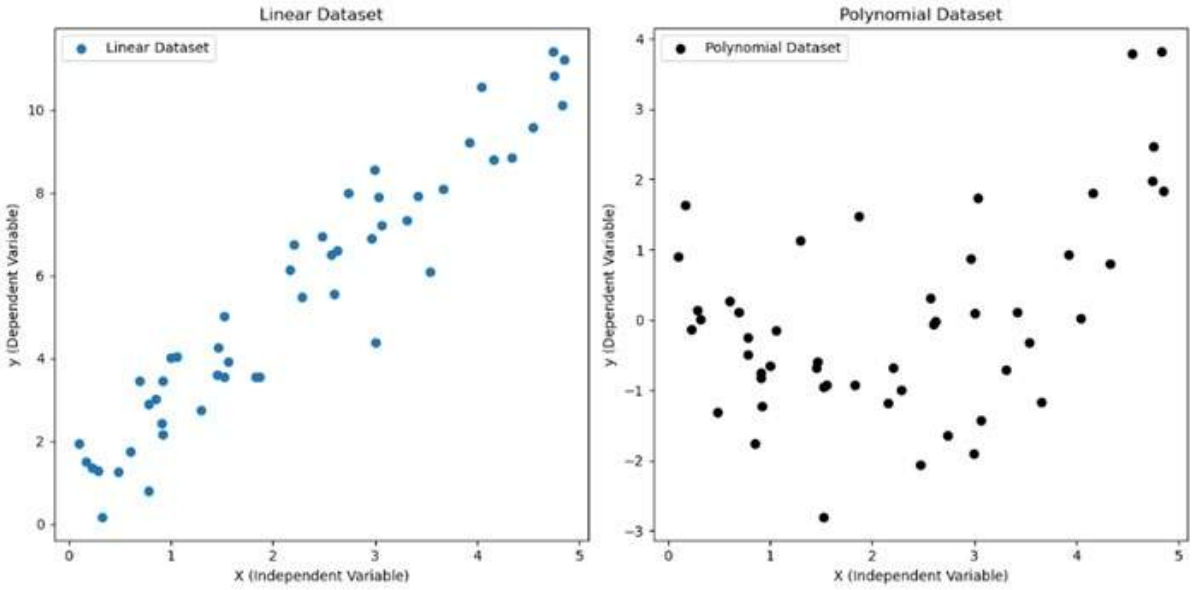


Figure 3.20. Linear vs polynomial datasets by Arden University.

From the datasets shown, using a linear regression line will not be suitable for both. So, how do we represent a polynomial regression line as an equation?

A non-linear (polynomial) regression line can be written as:

$$y = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$$

In the above, the *a*'s are the real numbers, *x* is what you want it to be, and the n's are the exponents. For example:

$$y = 4x^3 + 2x - 5$$

A polynomial that does have exponents is measured in degrees. So, in the example above, we have a polynomial of degree 3 because the exponent $4x^3$ is three. If the exponent had been 6, the polynomial would have been degree 6.

Continue

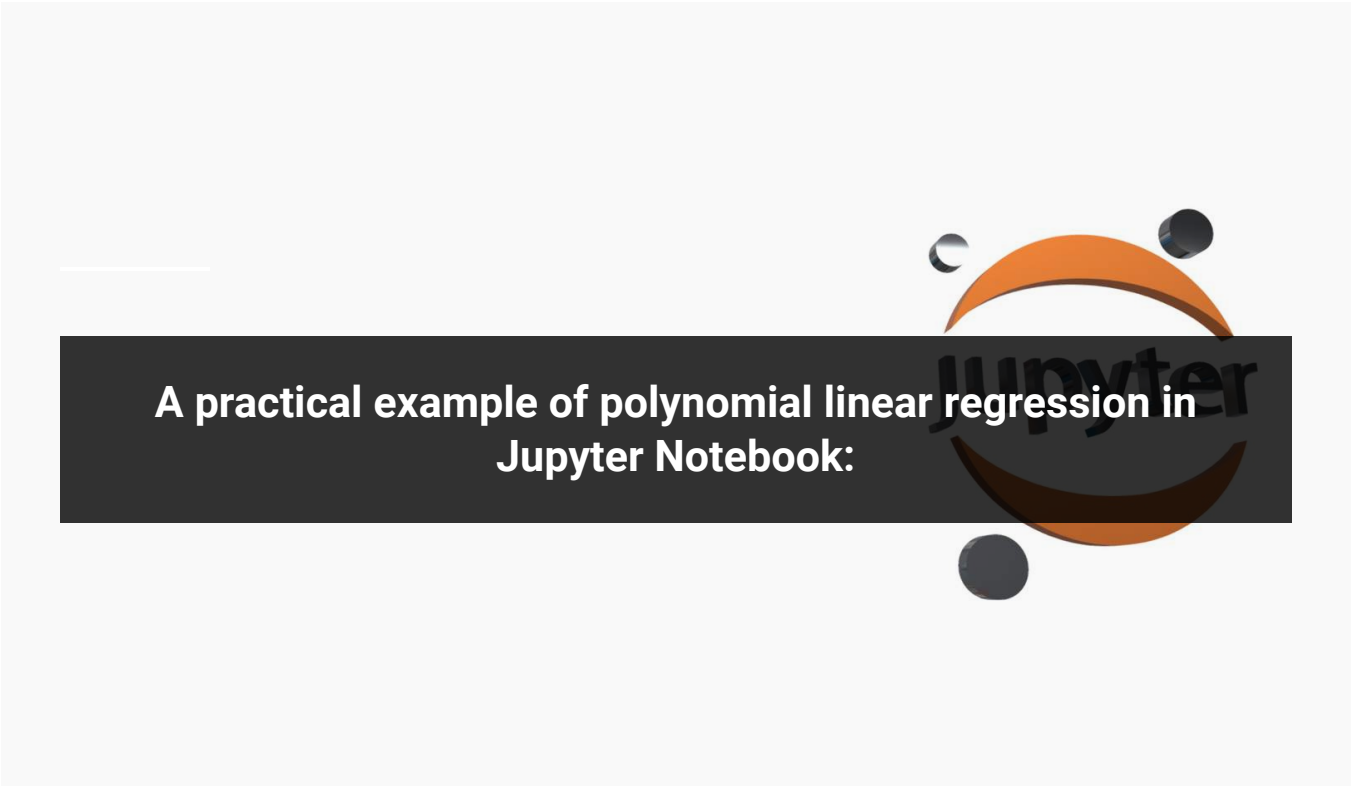


Image by wpswps from Sketchfab.com.

Hence, using the polynomial dataset in Figure 3.20, we can plot the regression line with the corresponding equation and R^2 using the following program code.

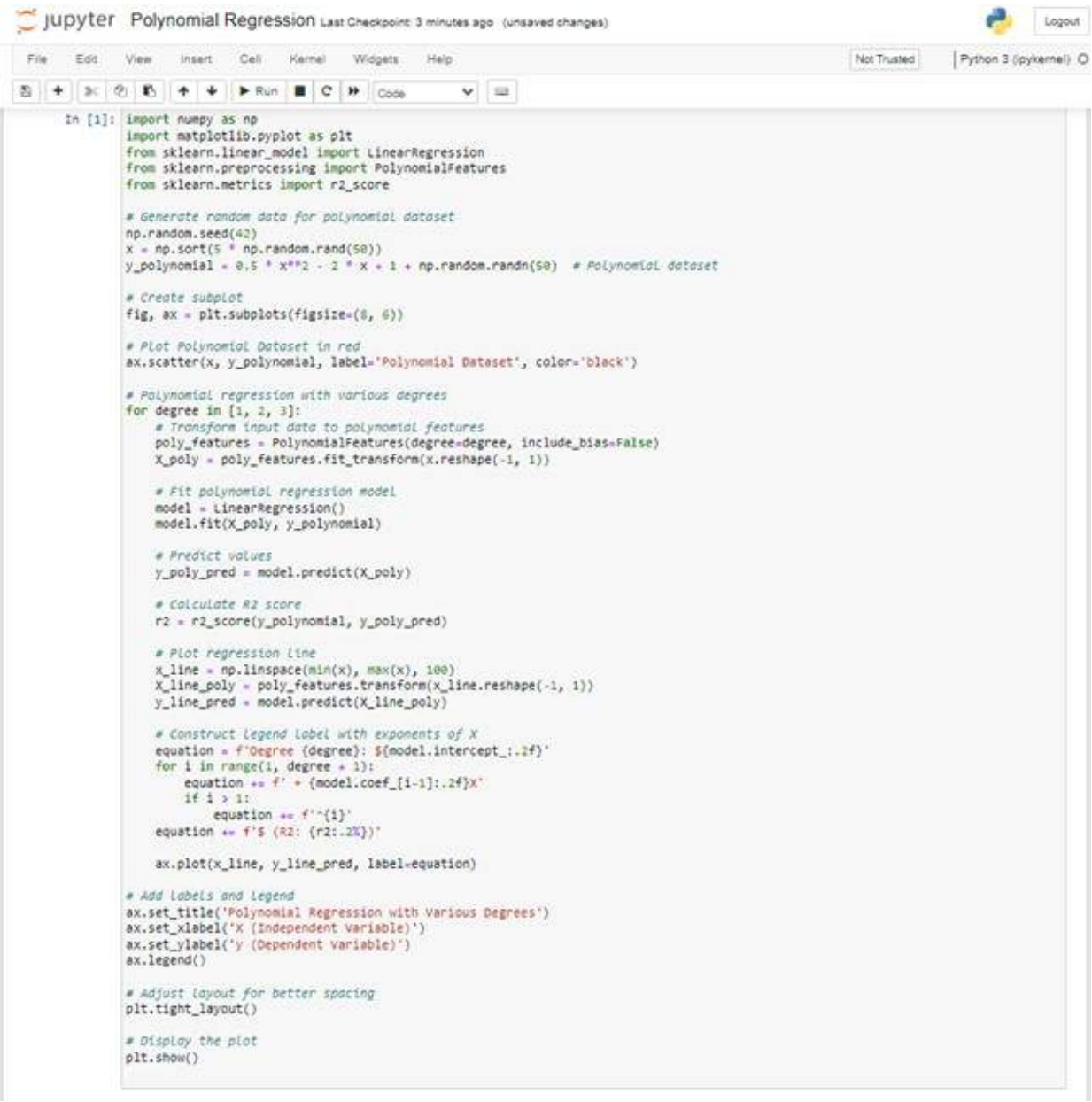


Figure 3.21. Polynomial regression model in Jupyter Notebook by Arden University. (Accessible code).

The output of the program is:

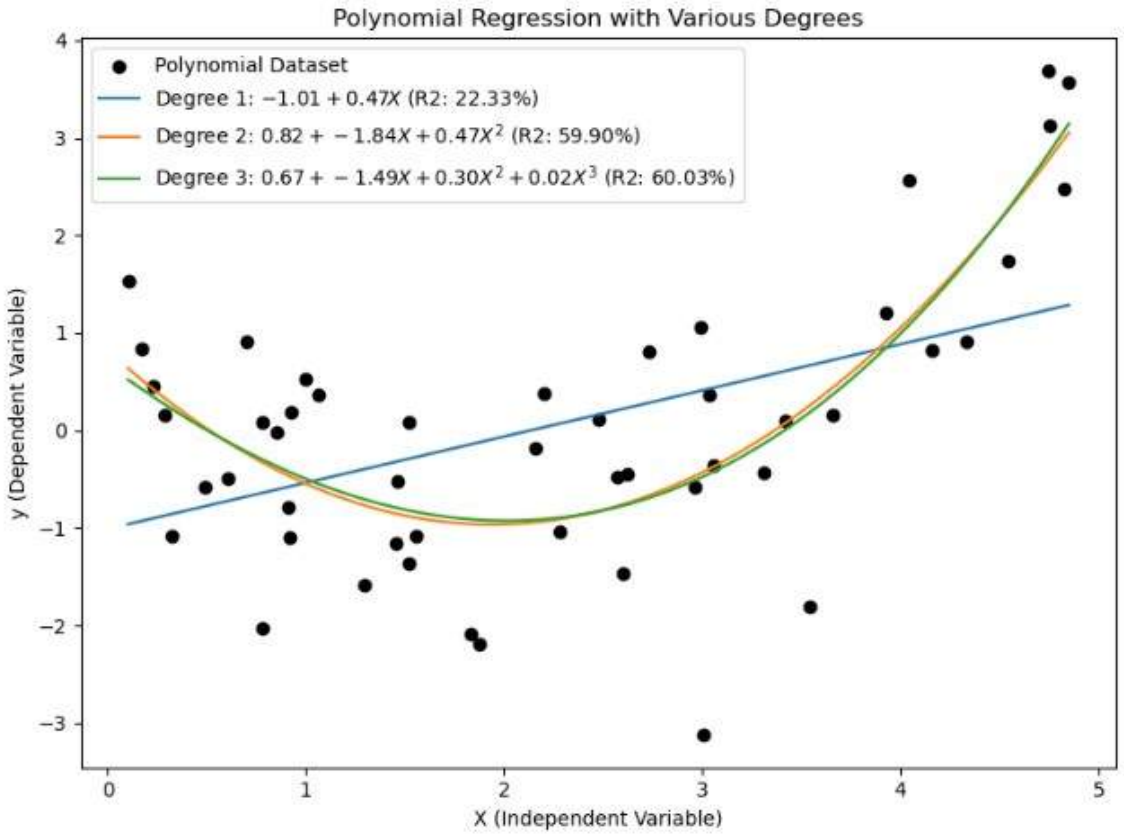


Figure 3.22. Polynomial regression model visualisation in Jupyter Notebook by Arden University. (Accessible code).

By examining the model's output, we know that degree 1 regression, a simple linear regression line, will not fit the data with R^2 of 22.33%. Therefore, we have increased the polynomial degree to degrees 2 and 3 to improve the model. As you can see, the level of improvement on the model for degrees 3 and 4 is not significant. Hence, we can say the performance has reached its equilibrium or is stabilised. Adding additional data points or higher degree regression can force a better performance. Still, we might run the risk of overfitting our model: creating a model that best fits our existing data point but is poor in predicting *y* for a new *x* value, as shown in the figure below:

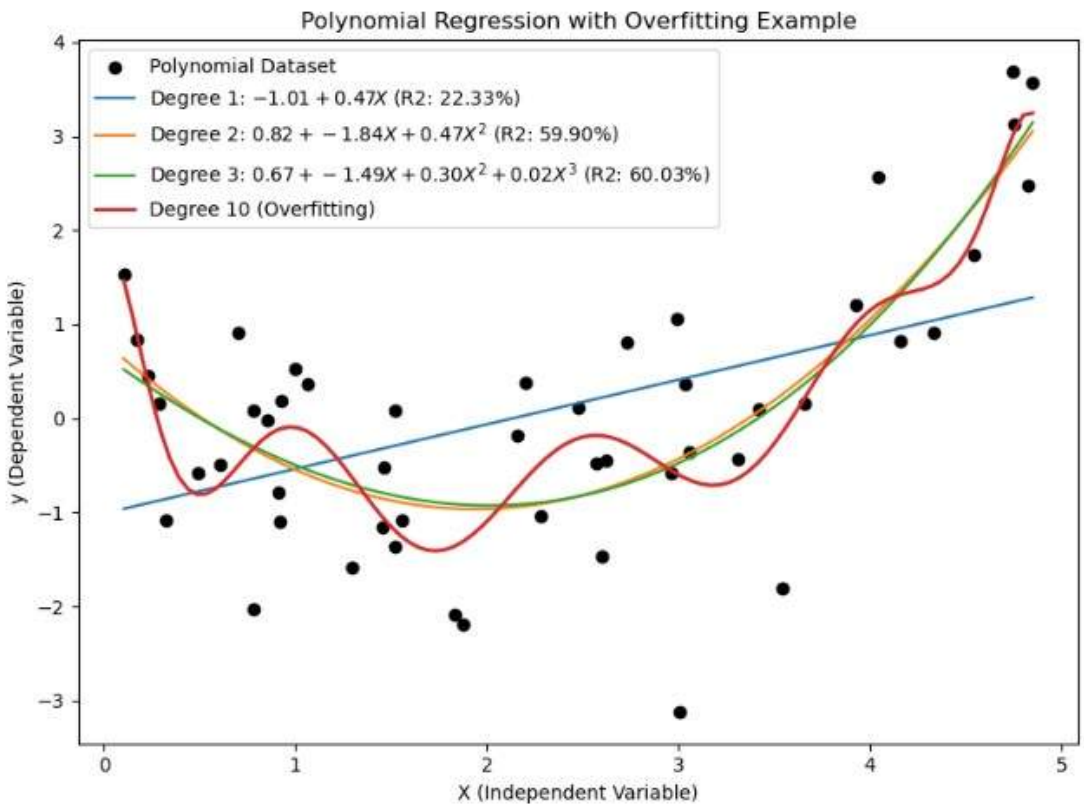


Figure 3.23. Polynomial regression model visualisation with overfitting in Jupyter Notebook by Arden University.

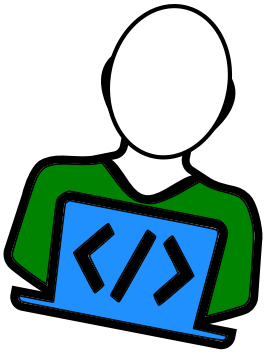
Continue

Topic 3 - Activity 1

Hands-on exercise to develop your polynomial regression model

Create a polynomial regression model using 100 randomly generated data points. Plot the line of best fit on the curve.

From the model created above, how could you improve its accuracy? Implement the improvements and run the model again. Have the changes improved the model's accuracy?



Continue

Topic 3 - Activity 2

Reading



- 13.10, 13.11 and 13.12 (pp. 382-395) from Chapter 13: Logistic Regression in Data Mining and Predictive Analytics by Larose and Larose

- Pages 124-130 of Chapter 4: Analysis of Business Intelligence and Big Data Adoption in Organizations from Business Intelligence and Big Data: Drivers of Organizational Success by Olszak

Continue

TOPIC REVIEW

In this lesson's first three topics, we explored the regression models, which are simple linear, multiple linear and polynomial regressions. Using the least mean squared (LMS) method, we can create the regression line and use R2 to measure the model's performance.

One important take from these topics is that when we engage with regression, we must first examine the correlation of the independent and dependent variables. This can be easily carried out by visualising them. In most cases, you will notice that they do not usually exhibit a linear trend but an exponential curve trend. Hence, polynomial regression allows a better fit.

Continue to Topic 4

Topic 4: LASSO regression

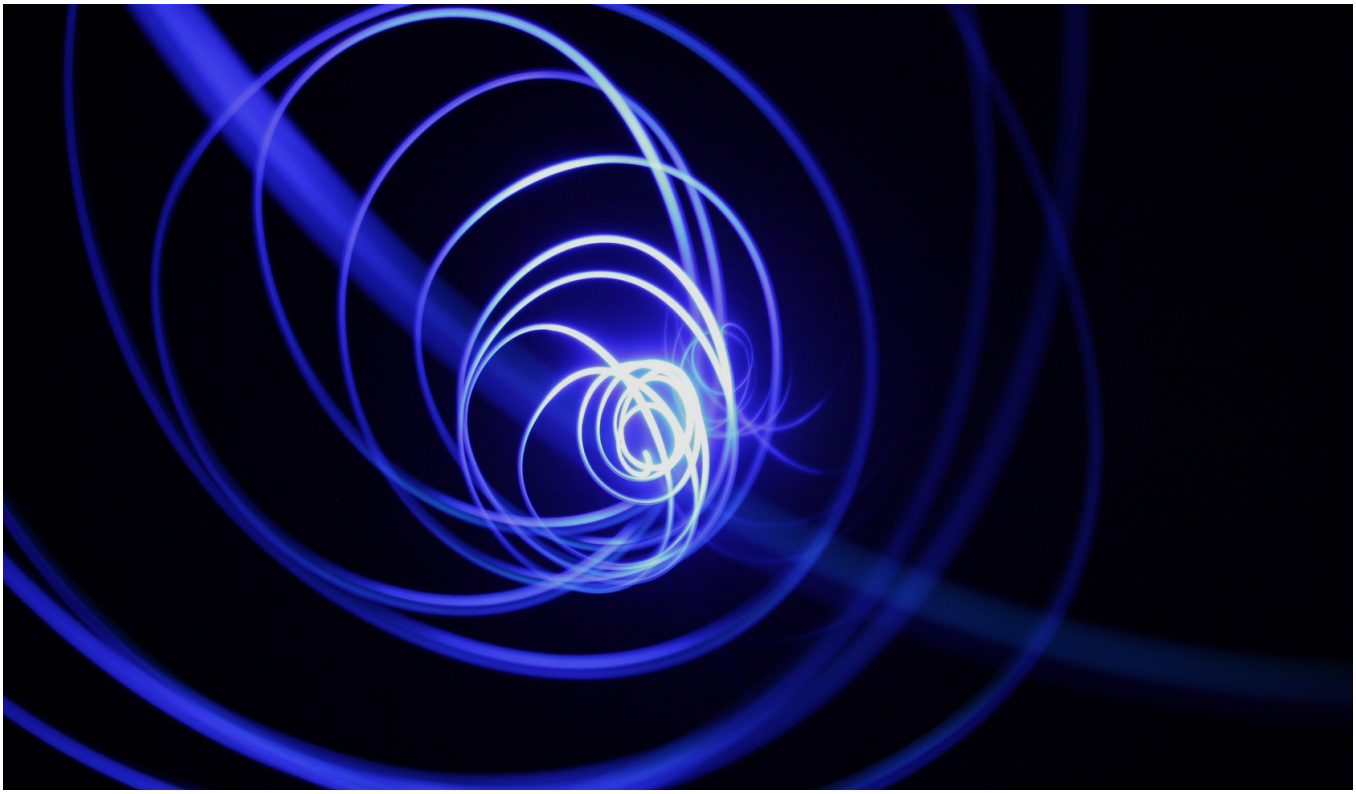


Image from Articulate Rise 360 Content Library.

When we deal with datasets with a high number of independent variables, in the case of MLR we use a backward elimination process to remove variables that do not contribute to the model's performance based on the *p*-value. In this topic, we will explore the least absolute shrinkage and selection operator (LASSO) regression, with an additional function of adding a penalty using L1 regularisation to linear regression, encouraging the model to use fewer independent variables. .

Why are fewer independent variables better?

Too many features can lead to overfitting (fitting the noise in the data), especially if some independent variables are highly correlated.

Continue



Image from Articulate Rise 360 Content Library.

LASSO regression employs the L1 regularisation technique, introducing penalties on the model's coefficients of the independent variables to reduce their number. As the magnitude of the penalties

increases, the coefficients tend to approach zero. When a coefficient reaches zero, it indicates that the corresponding independent variable is considered unimportant for prediction and can be excluded from the model.

A practical example of polynomial linear regression in Jupyter Notebook:

In the implementation of a LASSO model, we need to realise that multiple independent variables will limit our possibilities to visualise the final model. In the following example, we will be using the *california_housing_train.csv*, which is part of the sklearn package. Here is the program code to load and display the description and sample data from the dataset:

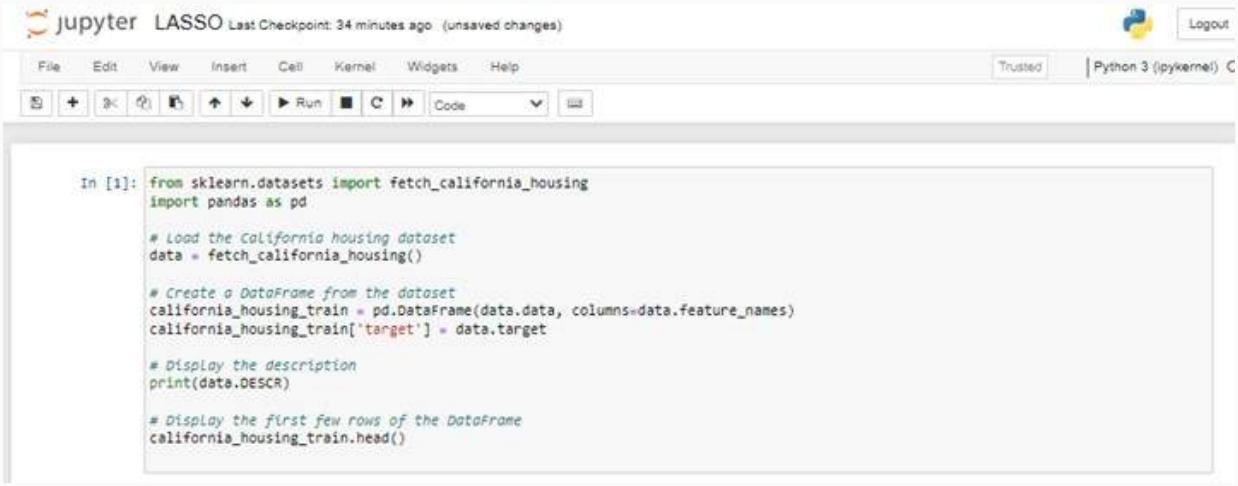


Figure 3.24. Loading dataset for LASSO regression model implementation in Jupyter Notebook by Arden University. (Accessible code).

The output:

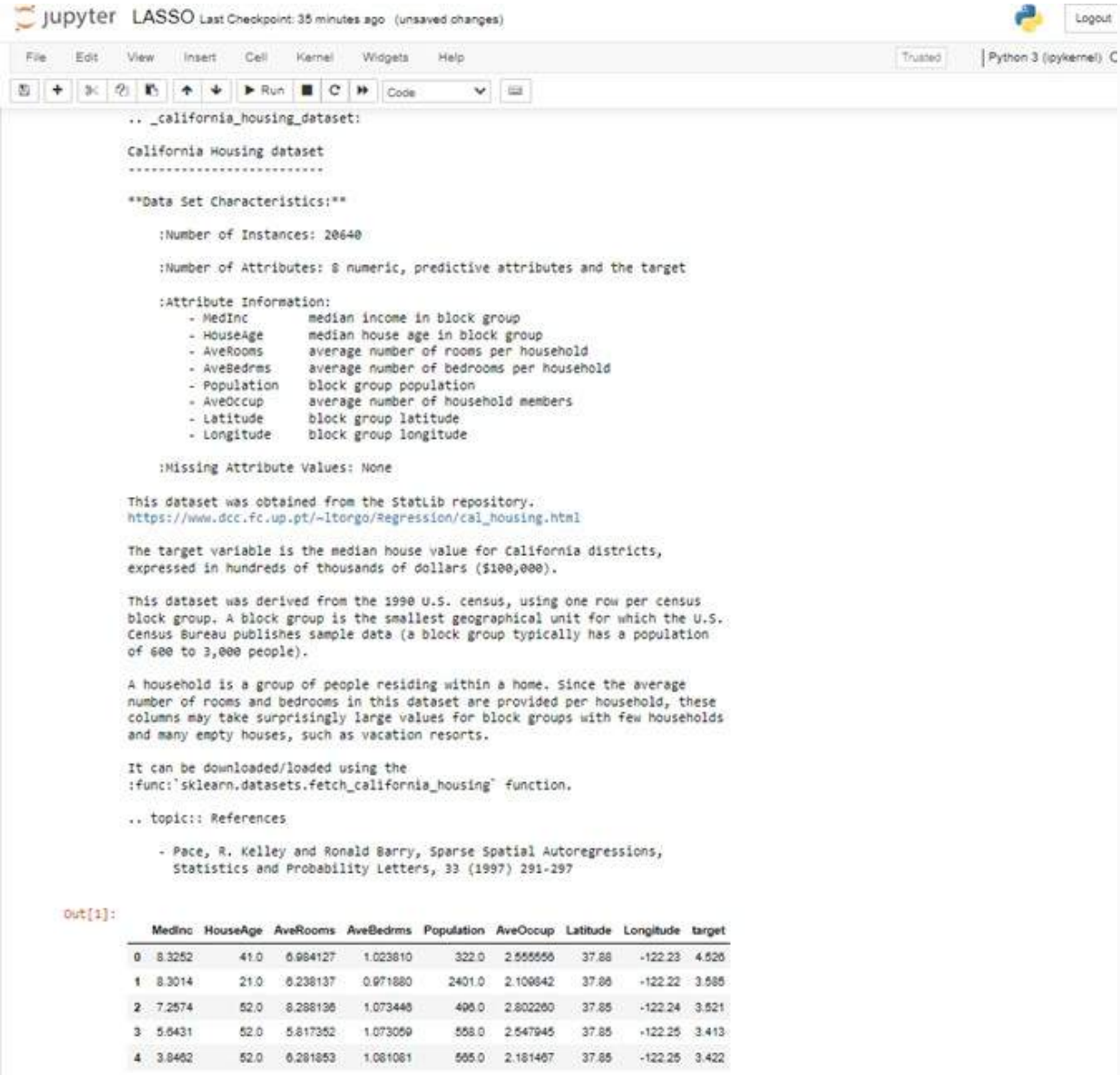


Figure 3.25. LASSO regression model implementation dataset description and samples in Jupyter Notebook by Arden University. (Accessible code).

One very useful feature when implementing LASSO regression is looking at the correlation between all the independent variables. This can be done using the following code:

```
# Calculate the correlation matrix
correlation_matrix = california_housing_train.corr()

# Create a heatmap of the correlation matrix
plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)
plt.title('Correlation Scatterplot Matrix')
plt.show()
```

Figure 3.26. LASSO regression model implementation variables correlation matrix in Jupyter Notebook by Arden University. (Accessible code).

The output:

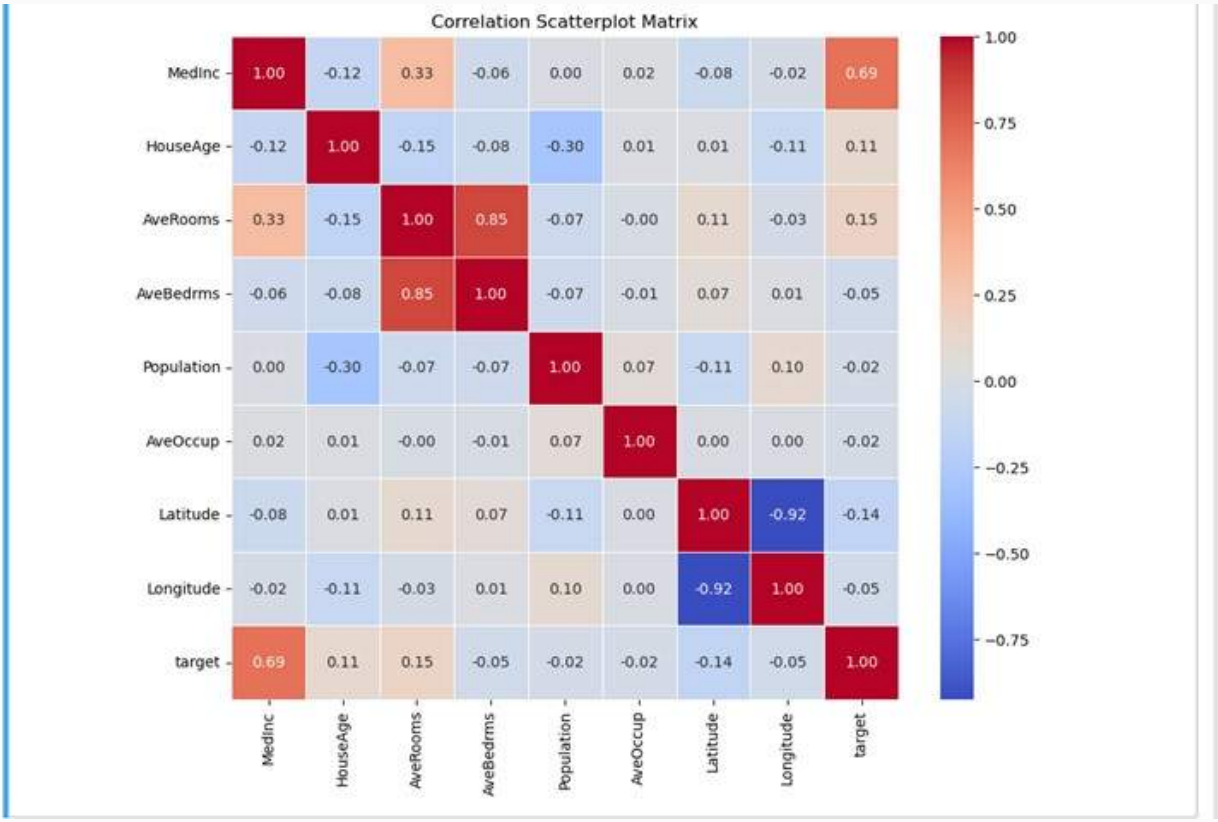


Figure 3.27. Independent variables correlation matrix in Jupyter Notebook by Arden University. (Accessible code).

Now let's implement our LASSO regression model:

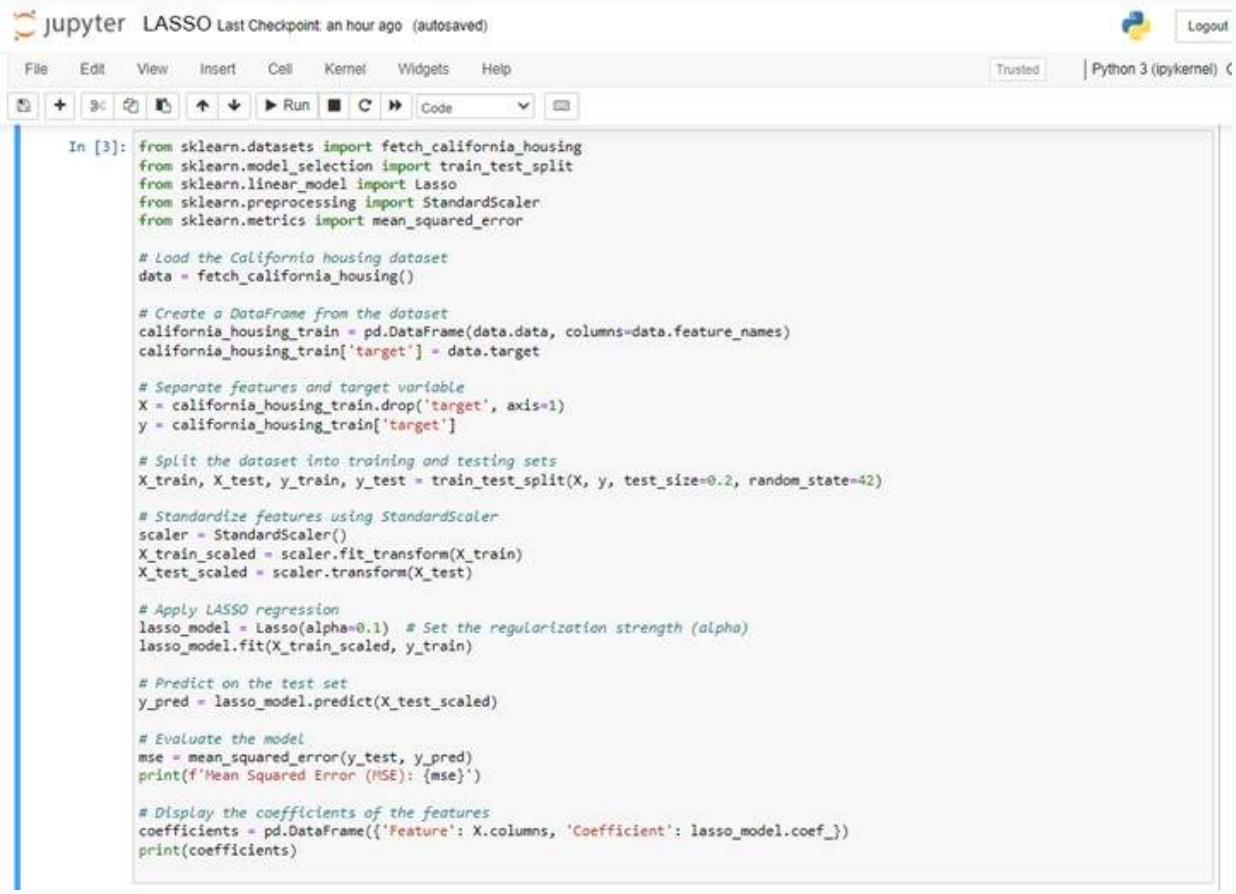


Figure 3.28. LASSO regression model in Jupyter Notebook by Arden University. (Accessible code).

The output:

```
Mean Squared Error (MSE): 0.6796290284328825
      Feature  Coefficient
0      MedInc     0.710598
1    HouseAge     0.106453
2    AveRooms    -0.000000
3   AveBedrms     0.000000
4  Population    -0.000000
5    AveOccup    -0.000000
6    Latitude    -0.011469
7   Longitude    -0.000000
```

Figure 3.29. LASSO regression model output in Jupyter Notebook by Arden University. (Accessible code).

The output of the LASSO regression code provides information about the model's performance and the coefficients of the independent variables or features. Here's what each part of the output generally tells us:

Mean squared error (MSE) is a measure that indicates how accurately a model's predictions match the actual values in the test set. A lower MSE means that the model has better predictive performance. The MSE represents the average squared difference between the predicted and actual median house values in the context of the California housing dataset.

The coefficients of the independent variables are the weights assigned to each feature by the LASSO regression model. In this dataset, positive coefficients indicate a positive correlation with the target variable (median house value), negative coefficients indicate a negative correlation, and coefficients close to zero suggest that the corresponding feature has minimal impact on the prediction.

By examining the coefficients, you will gain insights into the importance of each feature in predicting the target variable. LASSO's regularisation drives some coefficients to exactly zero,

which helps in performing feature selection. This is particularly useful when dealing with high-dimensional datasets, as it helps identify the most influential features and simplifies the model.

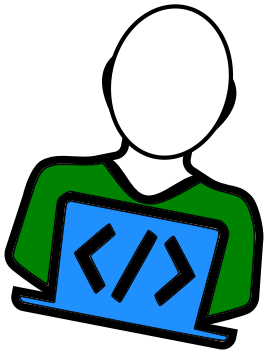
Continue

Topic 4 - Activity 1

Hands-on exercise to develop your own LASSO regression model

Create a LASSO regression model using 200 randomly generated data points with five features. How many features are important from the result?

From the model created above, how could you improve its accuracy? Implement the improvements and run the model again. Have the changes improved the model's accuracy?



Continue

Topic 4 - Activity 2

Reading



- Pages 130-136 of Chapter 4: Analysis of Business Intelligence and Big Data Adoption in Organizations in Business Intelligence and Big Data: Drivers of Organizational Success by Olszak
- Pages 147-167 of Chapter 4: Non-linear Functions and Applications in Essential Mathematics for Economics and Business by Bradley

Continue to Knowledge check

Knowledge check

Select the correct answer to the following questions.

1 of 3: What is the main idea for any simple linear regression analysis?

- The main idea is to make sure the scatter plot looks nice.
- The main idea is to use R^2 to fit a line to the data, determining a least squares value.
- The main idea is to use collinearity to fit a line to the data, determining an R^2 value.
- The main idea is to use least squares to fit a line to the data, determining an R^2 value.

SUBMIT

2 of 3: A polynomial is measured in degrees.

Is the following statement **true** or **false**?

A polynomial that has exponents is measured in degrees.

- ◯ True
- ◯ False

SUBMIT

3 of 3: Is the following statement **true** or **false**?

No degrees are required for it to be a polynomial.

- ◯ True
- ◯ False

SUBMIT

Continue to Lesson summary

Lesson summary

LESSON SUMMARY

Well done, we made it to the end of Lesson 3.

In this lesson, we have explored regression, from the intuition of linear regression to LASSO regression for complex, multiple-independent variables/features datasets.

Hopefully, now that we have gone through both the theory and several examples, you should have a better understanding of how to implement different types of regression models and, more importantly, a good technical knowledge that will enable you to identify the most suitable models based on the datasets and problem domains.

Continue to 'Reading list'

Reading List

All reading and resources in this module are provided by the Arden University Library service. The full reading list for this module can be accessed at the link below. More books, journals and databases can be found on the Library Portal. If you need these resources in a different format, our accessibility tools can help.

Full reading list

You can find all the key, essential and wider reading and resources for this module on the dedicated reading list page on the Learning Success Hub.

VIEW READING LIST

References

Cited sources that do not appear in your reading list:

McKinsey & Company, (2024). What is prompt engineering? *McKinsey & Company* [online], 22 March 2024. New York, NY: McKinsey & Company.